

Rollback Plan

Windows Detection Engineering Lab — Wazuh, Sysmon & MITRE ATT&CK

Project Name: Windows Detection Engineering Lab — Wazuh, Sysmon & MITRE ATT&CK

Document Type: Rollback Plan

Environment: Local VirtualBox Lab Environment

Primary Components: Wazuh Manager, Wazuh Dashboard, Windows 10 Endpoint, Sysmon, Windows Event Logs, Custom Wazuh Rules

1. Document Purpose

The purpose of this rollback plan is to define a safe, controlled, and repeatable process for removing, disabling, restoring, or rolling back custom Wazuh detection rules used in the **Windows Detection Engineering Lab — Wazuh, Sysmon & MITRE ATT&CK** project.

This document ensures that any custom detection logic added to the Wazuh manager can be safely reversed if it causes errors, false positives, dashboard issues, service instability, incorrect alert generation, XML syntax problems, or unintended detection behavior.

The rollback plan is designed to protect the stability of the Wazuh environment while preserving project evidence, screenshots, validation notes, and detection engineering documentation. It also demonstrates professional change-control thinking, which is important for real-world SOC, SIEM, and detection engineering workflows.

2. Rollback Objective

The main objective of this rollback plan is to restore the Wazuh environment to a known-good working state after custom rule changes are applied.

A successful rollback should achieve the following outcomes:

The Wazuh manager service should start successfully without XML parsing errors. The Wazuh dashboard should remain accessible and should continue displaying security events. Existing

default Wazuh rules should remain unaffected. Custom rule files should be safely removed, disabled, or restored from backup. Alerts generated by the Windows endpoint should continue to be processed correctly. Evidence related to the failed or reverted rule change should be preserved for documentation and analysis.

The rollback process should never delete original evidence, project screenshots, alert JSONs, detection notes, or validation documents unless they are explicitly identified as temporary test files.

3. Scope of This Rollback Plan

This rollback plan applies to custom Wazuh rule changes created for the Windows Detection Engineering Lab. It specifically covers custom rules created to detect Windows endpoint activity generated from Sysmon, Windows Security logs, Windows Defender, PowerShell, local user creation, Windows service creation or modification, and EICAR/Defender test events.

This rollback plan applies to the following systems:

Wazuh Server VM: Ubuntu 22.04 running Wazuh Manager, Wazuh Dashboard, indexer components, and custom rule files.

Windows 10 Lab Endpoint: Windows endpoint running Wazuh Agent, Sysmon, Windows Event Logs, and Windows Defender telemetry.

VirtualBox Host-Only Network: Local isolated lab network used to connect the Windows endpoint and Wazuh server.

Wazuh Dashboard: Web interface used for alert validation, MITRE ATT&CK mapping, screenshots, and detection review.

This rollback plan does not cover production systems, public IP targets, unauthorized machines, real malware testing, credential theft activity, ransomware simulation, command-and-control activity, or any activity outside the controlled local lab environment.

4. Lab Environment Summary

The lab environment consists of a Windows 10 endpoint forwarding telemetry to a Wazuh server through the Wazuh Agent. Sysmon is installed on the Windows endpoint to provide enhanced telemetry such as process creation, command-line execution, network activity, and system behavior. Wazuh receives and processes these logs using built-in decoders, default rules, and custom detection rules created for the project.

The Wazuh server is hosted on Ubuntu 22.04 inside VirtualBox. The Windows endpoint and Wazuh server communicate over a local host-only or internal lab network. The Wazuh dashboard is used to validate generated alerts, inspect rule matches, view MITRE ATT&CK mappings, and capture screenshots for project documentation.

5. Custom Rule Inventory

The following custom detections are part of the project scope and may require rollback if they cause instability, excessive false positives, or incorrect alert behavior.

Detection ID	Detection Name	Wazuh Rule ID	MITRE ATT&CK Mapping	Purpose
DET-001	PowerShell Encoded Command Detection	100100	T1059.001	Detects suspicious PowerShell encoded command execution
DET-002	Local User Creation Detection	100101	T1136.001	Detects creation of a local Windows user account
DET-003	Windows Service Creation / Modification	100102	T1543.003	Detects service creation or modification activity
DET-009	Defender / EICAR Test Detection	100108	T1204.002	Detects Windows Defender response to EICAR test file
DET-010	Additional Windows Detection Rule	100109	Project-specific	Detects additional Windows endpoint behavior based on project scope

The main custom rule file is expected to be located at:

```
/var/ossec/etc/rules/windows_detection_engineering_rules.xml
```

If the custom rules are stored under a different filename, the same rollback logic applies, but the filename must be updated in the commands.

6. Rollback Principles

Rollback should be performed carefully and in a controlled manner. The goal is not simply to delete the custom rule file, but to preserve evidence, identify the root cause, restore service stability, and document what happened.

The following rollback principles should always be followed:

First, always create a backup before modifying or removing rule files. Second, validate XML syntax and Wazuh configuration before restarting services. Third, never remove default Wazuh rules unless there is a specific documented reason. Fourth, preserve screenshots and alert evidence before making changes. Fifth, apply the smallest rollback action necessary. Sixth, confirm that Wazuh Manager, Dashboard, and endpoint alert forwarding are working after rollback. Seventh, document the rollback reason, commands used, validation result, and final status.

7. Rollback Triggers

Rollback should be initiated if any of the following conditions occur after deploying or modifying custom Wazuh rules:

The Wazuh manager fails to restart. The Wazuh manager logs show XML parsing errors. Custom rules generate excessive false positives. Alerts stop appearing in the Wazuh dashboard. The Windows endpoint appears disconnected after rule changes. The Wazuh dashboard becomes unstable or stops showing expected alerts. Detection logic matches the wrong events. MITRE ATT&CK mappings appear incorrect. Rule IDs conflict with existing rules. The rule file contains syntax errors, duplicate tags, unsupported fields, or invalid XML structure. Custom rules cause high noise during validation. A detection produces misleading results that could reduce the quality of the project.

Rollback should also be performed if the custom rule change was temporary and is no longer needed after validation.

8. Rollback Severity Levels

Rollback actions should be selected based on severity. Not every issue requires a full rollback.

Level 0 — No Rollback Required

This level applies when the custom rule is working correctly and only minor documentation changes are needed. No rule file changes are required. The analyst should update the detection notes, false-positive notes, or validation matrix only.

Level 1 — Rule Tuning Rollback

This level applies when the rule works but is too noisy or too broad. Instead of removing the rule, the detection logic should be tuned. Examples include lowering or raising the severity level, adding additional conditions, narrowing the matching pattern, or adding false-positive guidance.

Level 2 — Disable Specific Rule

This level applies when a specific custom detection is causing problems but the rest of the rule file is stable. The affected rule should be commented out or temporarily removed while leaving the remaining detections active.

Level 3 — Restore Previous Rule File

This level applies when multiple rules are affected or the updated custom rule file causes detection issues. The previous backup of the rule file should be restored.

Level 4 — Emergency Full Custom Rule Removal

This level applies when the Wazuh manager fails to start, the XML file is broken, or the rule file causes major instability. The custom rule file should be moved out of the active Wazuh rules directory and the Wazuh manager should be restarted.

9. Pre-Rollback Checklist

Before performing rollback, complete the following checks:

Confirm which custom rule change caused the issue. Identify the affected rule ID and detection ID. Capture screenshots of the error, alert behavior, or dashboard issue. Check the Wazuh manager service status. Review the Wazuh manager logs for XML or ruleset errors. Confirm whether the issue affects only one detection or the full rule file. Create a backup of the current rule file before changing it. Record the date, time, and reason for rollback.

Recommended evidence to capture before rollback includes Wazuh dashboard screenshots, alert JSON, rule file screenshots, manager log output, service status output, and validation notes.

10. Backup Procedure Before Rollback

Before modifying or removing the custom rule file, create a timestamped backup.

Run the following command on the **Ubuntu Wazuh Server VM**:

```
sudo cp /var/ossec/etc/rules/windows_detection_engineering_rules.xml  
/var/ossec/etc/rules/windows_detection_engineering_rules.xml.backup_$(date  
+%Y%m%d_%H%M%S)
```

Verify that the backup was created:

```
ls -l /var/ossec/etc/rules/ | grep windows_detection_engineering_rules
```

Create an additional backup in the project evidence directory if required:

```
mkdir -p ~/wazuh-rule-backups  
sudo cp /var/ossec/etc/rules/windows_detection_engineering_rules.xml ~/wazuh-rule-backups/
```

This ensures that even if the active rule file is removed from Wazuh, the project retains a copy for analysis and documentation.

11. Check Wazuh Manager Status

Before rollback, check whether the Wazuh manager is running.

Run this on the **Ubuntu Wazuh Server VM**:

```
sudo systemctl status wazuh-manager
```

If the service is active and running, the issue may be limited to rule behavior or alert noise. If the service is failed, the issue may be related to XML syntax, rule conflict, invalid configuration, or unsupported rule structure.

To view recent manager logs, run:

```
sudo tail -n 50 /var/ossec/logs/ossec.log
```

To search for rule-related errors, run:

```
sudo grep -i "error" /var/ossec/logs/ossec.log | tail -n 30
```

Also check for XML or rule parsing messages:

```
sudo grep -i "rules" /var/ossec/logs/ossec.log | tail -n 30
```

12. XML Validation Before Restart

Before restarting Wazuh after any rollback action, validate the XML structure of the custom rule file.

If `xmllint` is installed, run:

```
xmllint --noout /var/ossec/etc/rules/windows_detection_engineering_rules.xml
```

If `xmllint` is not installed, install it:

```
sudo apt update  
sudo apt install libxml2-utils -y
```

Then run the validation again:

```
xmllint --noout /var/ossec/etc/rules/windows_detection_engineering_rules.xml
```

A successful result should return no output. If there is an error, the command will show the line number and issue.

Common XML problems include missing closing tags, extra content after the final closing tag, invalid characters, duplicate XML root elements, unescaped symbols, and incorrectly nested rule elements.

13. Rollback Procedure — Level 1: Tune a Noisy Rule

Use this rollback level when the rule is generating alerts but the alerts are too noisy or too broad.

Examples include a PowerShell rule triggering on normal administrative activity, a service creation rule alerting on legitimate Windows services, or a local user rule triggering during normal account management.

Steps:

Open the custom Wazuh rule file:

```
sudo nano /var/ossec/etc/rules/windows_detection_engineering_rules.xml
```

Locate the noisy rule by rule ID. For example, locate rule `100100` for PowerShell Encoded Command detection.

Adjust the rule to make it more specific. This may include adding additional match conditions, narrowing the event source, changing the severity level, or improving the rule description.

After editing, save the file.

Validate XML:

```
xmllint --noout /var/ossec/etc/rules/windows_detection_engineering_rules.xml
```

Restart Wazuh manager:

```
sudo systemctl restart wazuh-manager
```

Confirm service status:

```
sudo systemctl status wazuh-manager
```

Monitor logs:

```
sudo tail -f /var/ossec/logs/ossec.log
```

Validate in the Wazuh dashboard that the noisy alerts have been reduced.

14. Rollback Procedure — Level 2: Disable One Specific Custom Rule

Use this rollback level when only one detection rule is causing issues.

For example, if **DET-001** PowerShell Encoded Command detection is too noisy but the other rules are working, disable only rule **100100**.

Open the rule file:

```
sudo nano /var/ossec/etc/rules/windows_detection_engineering_rules.xml
```

Find the affected rule block. Example:

```
<rule id="100100" level="10">
  ...
</rule>
```

Comment out the full rule block using XML comments:

```
<!--
<rule id="100100" level="10">
  ...
</rule>
```


-->

Save the file.

Validate XML:

```
xmllint --noout /var/ossec/etc/rules/windows_detection_engineering_rules.xml
```

Restart the Wazuh manager:

```
sudo systemctl restart wazuh-manager
```

Check service status:

```
sudo systemctl status wazuh-manager
```

Check logs:

```
sudo tail -n 50 /var/ossec/logs/ossec.log
```

Confirm in the Wazuh dashboard that the disabled rule no longer generates alerts while other custom detections continue working.

Document the disabled rule in the rollback log.

15. Rollback Procedure — Level 3: Restore Previous Rule File Backup

Use this rollback level when multiple rules are affected, a new version of the rule file behaves incorrectly, or the previous version was stable and should be restored.

List available backups:

```
ls -lh /var/ossec/etc/rules/ | grep windows_detection_engineering_rules
```

Move the current problematic file to a failed-change copy:

```
sudo mv /var/ossec/etc/rules/windows_detection_engineering_rules.xml  
/var/ossec/etc/rules/windows_detection_engineering_rules.xml.failed_$(date  
+%Y%m%d_%H%M%S)
```

Restore the known-good backup:

```
sudo cp
/var/ossec/etc/rules/windows_detection_engineering_rules.xml.backup_YYYYMMDD_HHMMSS
/var/ossec/etc/rules/windows_detection_engineering_rules.xml
```

Replace `YYYYMMDD_HHMMSS` with the correct timestamp from the backup file.

Validate the restored XML file:

```
xmllint --noout /var/ossec/etc/rules/windows_detection_engineering_rules.xml
```

Restart Wazuh manager:

```
sudo systemctl restart wazuh-manager
```

Check service status:

```
sudo systemctl status wazuh-manager
```

Check logs:

```
sudo tail -n 50 /var/ossec/logs/ossec.log
```

Verify that the Wazuh dashboard is accessible and that alerts are being processed normally.

16. Rollback Procedure — Level 4: Emergency Removal of Custom Rule File

Use this rollback level only if the Wazuh manager fails to start or the custom rule file is causing serious instability.

Move the custom rule file out of the active rules directory:

```
sudo mv /var/ossec/etc/rules/windows_detection_engineering_rules.xml
/var/ossec/etc/rules/windows_detection_engineering_rules.xml.disabled_$(date
+%Y%m%d_%H%M%S)
```

Restart Wazuh manager:

```
sudo systemctl restart wazuh-manager
```

Check service status:

```
sudo systemctl status wazuh-manager
```

If the service starts successfully, confirm the issue was caused by the custom rule file.

Check logs:

```
sudo tail -n 50 /var/ossec/logs/ossec.log
```

Open the Wazuh dashboard and verify that the system is accessible.

This emergency rollback removes only the custom project rule file from active processing. It does not remove default Wazuh rules, agent configuration, Sysmon configuration, or previous evidence.

17. Rule-Specific Rollback Guidance

DET-001 — PowerShell Encoded Command Detection

This rule is used to detect suspicious PowerShell encoded command behavior and maps to MITRE ATT&CK technique [T1059.001](#).

Rollback may be required if the rule triggers on normal PowerShell administration, legitimate scripts, system management tools, or expected lab commands.

Recommended rollback action is to tune the command-line match condition before fully disabling the rule. The rule should focus on high-risk indicators such as [EncodedCommand](#), [-enc](#), suspicious execution policy bypass, or encoded PowerShell patterns.

If the rule remains too noisy, disable only rule [100100](#) while keeping the rest of the custom rule file active.

DET-002 — Local User Creation Detection

This rule is used to detect local Windows account creation and maps to MITRE ATT&CK technique [T1136.001](#).

Rollback may be required if the rule triggers during legitimate account creation, Windows setup activity, administrator testing, or lab reset procedures.

Recommended rollback action is to add context to the rule or document expected false positives. Since local user creation is a meaningful detection in a SOC environment, full removal should only be performed if the rule is technically broken or mapped incorrectly.

If needed, disable only rule 100101.

DET-003 — Windows Service Creation / Modification Detection

This rule is used to detect Windows service creation or modification and maps to MITRE ATT&CK technique T1543.003.

Rollback may be required if the rule triggers on legitimate software installations, Windows updates, driver installations, or endpoint security tool activity.

Recommended rollback action is to tune the rule by adding exclusions for known legitimate services or by reducing severity during testing. Service creation can be both legitimate and suspicious, so the detection should include false-positive notes.

If the rule causes excessive alert noise, disable only rule 100102.

DET-009 — Defender / EICAR Detection

This rule is used to detect Windows Defender activity related to EICAR test file detection and maps to MITRE ATT&CK technique T1204.002.

Rollback may be required if the rule incorrectly matches unrelated Defender events or if the detection logic is too broad.

Recommended rollback action is to tune the rule to match EICAR-specific indicators, Defender detection names, or expected Windows Defender event fields.

If the rule is not needed after validation, disable rule 100108 but retain evidence screenshots and alert JSONs for the project report.

DET-010 — Additional Windows Detection Rule

This rule is project-specific and should be rolled back based on its detection objective.

Rollback may be required if the rule duplicates an existing Wazuh rule, causes rule ID conflicts, produces incorrect MITRE mappings, or generates poor-quality alerts.

Recommended rollback action is to verify whether the rule adds unique detection value. If it overlaps with another rule, either tune it or disable rule 100109.

18. Post-Rollback Validation

After rollback, perform validation to confirm that Wazuh is stable and the environment is functioning correctly.

Check Wazuh manager status:

```
sudo systemctl status wazuh-manager
```

Check recent logs:

```
sudo tail -n 50 /var/ossec/logs/ossec.log
```

Confirm that there are no XML errors, ruleset errors, or manager startup failures.

Open the Wazuh dashboard in the browser:

<https://<wazuh-server-ip>>

Confirm that the dashboard loads successfully.

Check that the Windows endpoint still appears as active.

Generate a simple test event from the Windows endpoint if needed. For example, run a harmless command such as PowerShell or `whoami` on the Windows VM and confirm that telemetry continues flowing.

Confirm the following:

The Wazuh manager is running. The Wazuh dashboard is accessible. The Windows agent is connected. Default Wazuh rules still generate alerts. Disabled custom rules no longer generate alerts. Unaffected custom rules still work if they were not rolled back. No new XML or service errors appear in logs.

19. Evidence Preservation During Rollback

Rollback should not remove project evidence. The following items should be preserved:

Screenshots of alerts, Wazuh dashboard views, MITRE ATT&CK mapping, validation matrix, event JSON files, detection notes, analyst triage guide, screenshots showing false positives, and logs showing rollback reasons.

If a rule file is removed or disabled, the old version should be saved with a timestamp instead of deleted.

Recommended naming format:

`windows_detection_engineering_rules.xml.failed_YYYYMMDD_HHMMSS`

or:

`windows_detection_engineering_rules.xml.disabled_YYYYMMDD_HHMMSS`

This preserves the failed rule version for future analysis.

20. Rollback Documentation Template

Every rollback should be documented using the following format.

Rollback ID: RB-[Number]

Date and Time: [Insert Date and Time]

Performed By: [Insert Name]

Affected Detection: [DET-001 / DET-002 / DET-003 / DET-009 / DET-010]

Affected Rule ID: [100100 / 100101 / 100102 / 100108 / 100109]

Rollback Level: [Level 1 / Level 2 / Level 3 / Level 4]

Reason for Rollback: [Describe issue]

Symptoms Observed: [Service failure, XML error, false positives, incorrect alerting, etc.]

Backup Created: [Yes / No]

Backup Filename: [Insert filename]

Commands Executed: [List commands]

Validation Result: [Passed / Failed]

Final Status: [Stable / Requires further tuning]

Evidence Preserved: [Screenshots, JSON, logs, etc.]

Notes: [Additional analyst notes]

21. Example Rollback Log Entry

Rollback ID: RB-001

Date and Time: [Insert Date]

Performed By: Rimanshu Sharma

Affected Detection: DET-001 — PowerShell Encoded Command Detection

Affected Rule ID: 100100

Rollback Level: Level 2 — Disable Specific Rule

Reason for Rollback: The rule generated excessive alerts during normal PowerShell testing activity.

Symptoms Observed: Multiple repeated PowerShell alerts appeared in the Wazuh dashboard during controlled lab validation.

Backup Created: Yes

Backup Filename: windows_detection_engineering_rules.xml.backup_20260514_153000

Commands Executed: Backup rule file, commented out rule 100100, validated XML, restarted Wazuh manager.

Validation Result: Passed

Final Status: Wazuh manager running, dashboard accessible, other custom detections working.

Evidence Preserved: Alert screenshots, detection notes, rule backup, manager log output.

Notes: Rule should be tuned with stricter command-line matching before being re-enabled.

22. Change-Control Checklist

Before applying or rolling back custom Wazuh rules, complete the following checklist:

Check	Status
Current rule file backed up	Pending / Complete
Affected rule ID identified	Pending / Complete
Affected detection ID identified	Pending / Complete
Wazuh manager status checked	Pending / Complete
XML syntax validated	Pending / Complete
Screenshots captured	Pending / Complete
Rollback reason documented	Pending / Complete

Wazuh manager restarted successfully	Pending / Complete
Wazuh dashboard verified	Pending / Complete
Windows agent status verified	Pending / Complete
Post-rollback alerts reviewed	Pending / Complete
Final rollback notes completed	Pending / Complete

23. Safe Rollback Command Set

The following commands provide a quick reference for safe rollback.

Create backup:

```
sudo cp /var/ossec/etc/rules/windows_detection_engineering_rules.xml
/var/ossec/etc/rules/windows_detection_engineering_rules.xml.backup_$(date
+%Y%m%d_%H%M%S)
```

Validate XML:

```
xmllint --noout /var/ossec/etc/rules/windows_detection_engineering_rules.xml
```

Edit custom rule file:

```
sudo nano /var/ossec/etc/rules/windows_detection_engineering_rules.xml
```

Restart Wazuh manager:

```
sudo systemctl restart wazuh-manager
```

Check Wazuh manager status:

```
sudo systemctl status wazuh-manager
```

View Wazuh logs:

```
sudo tail -n 50 /var/ossec/logs/ossec.log
```


Emergency disable custom rule file:

```
sudo mv /var/ossec/etc/rules/windows_detection_engineering_rules.xml  
/var/ossec/etc/rules/windows_detection_engineering_rules.xml.disabled_$(date  
+%Y%m%d_%H%M%S)
```

Restart after emergency disable:

```
sudo systemctl restart wazuh-manager
```

Confirm no errors:

```
sudo grep -i "error" /var/ossec/logs/ossec.log | tail -n 30
```

24. Rollback Success Criteria

Rollback is considered successful when all of the following conditions are met:

The Wazuh manager service is active and running. The Wazuh dashboard is accessible. The Windows endpoint remains connected. No XML parsing errors appear in Wazuh logs. No rule conflict errors appear in Wazuh logs. The problematic custom rule is removed, disabled, restored, or tuned. Default Wazuh detections remain unaffected. Project evidence is preserved. The rollback action is documented. The environment is stable enough for further validation or testing.

25. Risks and Mitigations

Risk	Impact	Mitigation
Deleting the only copy of the custom rule file	Loss of project work	Always create timestamped backups
XML syntax error after manual editing	Wazuh manager may fail to restart	Validate XML before restart
Disabling all custom rules accidentally	Loss of detection coverage	Disable only affected rule when possible
Removing evidence during rollback	Project documentation becomes incomplete	Preserve screenshots, JSON, and logs

Restarting Wazuh without validation	Service outage or repeated failure	Validate XML and check logs first
Rule ID conflict	Incorrect alert behavior	Use unique rule IDs and verify before deployment
False positives remain after rollback	Poor detection quality	Add false-positive notes and tune detection logic

26. Professional Notes for Portfolio Presentation

This rollback plan demonstrates that the project was not only focused on creating detections, but also on safely managing detection changes. In real SOC and detection engineering environments, rollback planning is important because even a small SIEM rule change can create alert noise, break parsing, reduce analyst trust, or affect monitoring reliability.

By including a rollback plan, the project demonstrates professional awareness of change control, operational stability, detection lifecycle management, false-positive handling, evidence preservation, and responsible security testing.

This strengthens the overall quality of the **Windows Detection Engineering Lab — Wazuh, Sysmon & MITRE ATT&CK** project because it shows the ability to build, test, validate, document, and safely reverse custom detection logic.

27. Conclusion

This rollback plan provides a structured and professional process for safely removing, disabling, restoring, or tuning custom Wazuh rules used in the Windows Detection Engineering Lab. It ensures that custom detection development can be performed without risking long-term instability of the Wazuh lab environment.